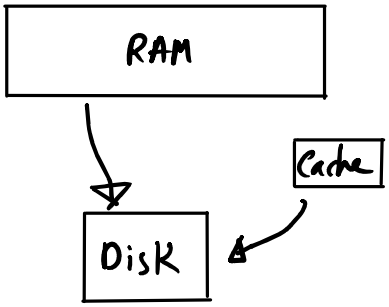
Problem solving using STL

Thursday, April 2, 2026 7:34 PM

LRU Cache — Least Recently Used



[D5, D6, D3, D4]

get(D3)

LFU Cache

① get (Key) → value

② put (Key, value)

map

Storage →

Queue ✗

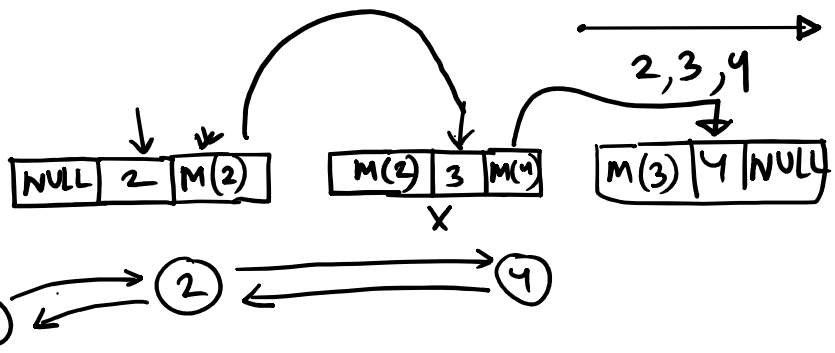
Stack ✗

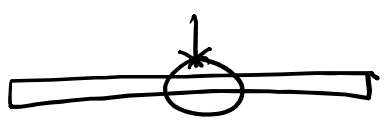
vector ✓

Doubly LinkedList

O(1)

put(1, 10)
put(2, 11)
put(3, 13)
put(4, 14)
get(3)





DLL

capacity = 3

Key Value (HashMap)

put(1, 10) ✓
put(2, 11) ✓
put(3, 13) ✓
put(4, 14) ✓
get(3) ✓
get(4) ✓
put(5, 16) ✓

3, 4, 5

[1, 2, 10, 7, 6]

get(10) [1, 2, 7, 6, 10]

get(10) [1, 2, 7, 6, 10]

get(2) [1, 7, 6, 10, 2]

address[1] = M1

address[2] = M2

→ address[10] = M10

address[7] = M7

400

401

mp[3] = 13

mp[4] = 14

mp[5] = 16

1, 2

~~2~~, 1

→ address[10] = M_{10}
address[7] = M_7
address[6] = M_6

401 ~~2~~, 1
402 1, 3

e → a
g → d

a ↔ c
b ↔ d

egg → add

saa

nqv

e → x
g → q

e x g z
↓ ↓
a b

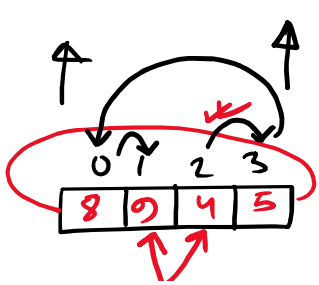
e n
↓ ↓
a c

cc
a a
↓ ↓
c d

dd
b b
↓ ↓
d c

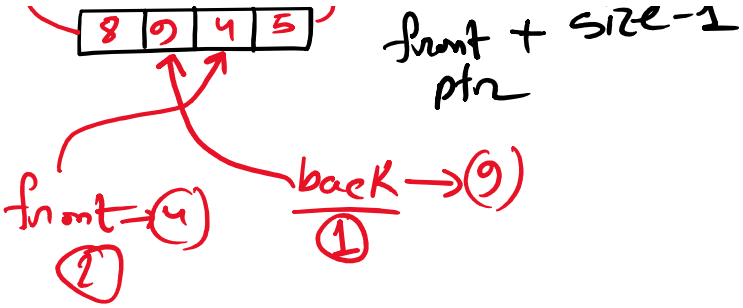
0 1 2 3 4 5 6
s = e g b b e g z
t = a d n n a d a

s t
e ⇄ a
g ⇄ d
b ⇄ n



size = 4

front + size − 1
ptr


8
9
4
5
front + size-1
ptr
front → 4
2
back → 9
1